PRODUCT REQUIREMENT DOCUMENT (PRD)

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Project Name: Z.Clear

Version: 2.0

1. Project Overview

- **Problem Statement:** Across Malaysia and globally, many workflows remain manual, fragmented, and inefficient, requiring constant human coordination and decision-making. Over 85% of trade documents (invoices, packing lists, BOLs) are unstructured text or PDFs. Manual data entry and verification are error-prone and slow, especially in multi-language environments like Malaysia. Customs regulations and HS Codes change frequently, making it difficult for SMEs to stay compliant in real-time
- **Target Domain:** Building an AI-powered system using GLM to transform inefficient workflows into intelligent, automated processes.
- **Proposed Solution Summary:** A sovereign, autonomous middleware designed to transform unstructured trade documents into actionable, stateful, and compliant data streams using Multi-Agent workflows.

2. Background & Business Objective

- **Background of the Problem:** Small and Medium-sized Enterprises (SMEs) are the backbone of global trade, yet they face disproportionate burdens in customs compliance. Unlike large corporations with dedicated legal departments, SMEs often struggle with the high volatility of international regulations, where rules regarding tariffs, restricted goods, and HS code classifications change frequently. Manual document processing leads to significant human error, resulting in missing or outdated paperwork, incorrect formats, and compliance gaps. These inefficiencies cause costly shipment delays, heavy administrative fines, and "border blockages."
- **Importance of Solving This Issue:** By automating regulatory tracking and document validation, this AI system reduces "border friction", potentially cutting shipment clearance

times. Most importantly, it democratizes global trade by providing small businesses with the same "regulatory intelligence" as multi-national corporations. The system allows SMEs to expand internationally without the prohibitive costs of hiring specialized legal and logistics teams.

- **Strategic Fit / Impact:** This AI system strategically transitions SMEs into the "TradeTech" era by automating complex regulatory hurdles. The impact is a significant reduction in operational costs and legal risks, effectively leveling the global playing field and allowing small businesses to scale internationally with minimal administrative overhead.

3. Product Purpose

3.1. Main Goal of the System: To automate and standardize the extraction of critical trade and logistics data from unstructured documents, ensuring that information required for customs clearance is captured accurately, efficiently, and reliably.

3.2. Intended Users (Target Audience):

- Small and Medium-sized Enterprises (SMEs) Logistic Provider
- Freight Forwarder
- Customs Brokers
- Cross-border E-commerce Sellers

4. System Functionalities

4.1. Description: The core objective of the system is to automate the extraction and structuring of customs-related data from unstructured text inputs. It converts free-form content—such as emails, chat messages, and informal notes—into standardized, relational data that can be directly

integrated into an SME's logistics database. By reducing manual data entry, the system improves accuracy, processing speed, and operational efficiency in custom documentation workflows.

4.2. Key Functionalities:

- **Robust Handling of Unstructured Input:** Automatically identifies and parses trade data from messy email bodies, informal chat logs, or shorthand notes, regardless of inconsistent formatting or non-standard layouts.
- **One-Click Key Data Retrieval:** Enables single-click extraction of essential customs clearance information, such as **Goods Description, HS Code, and Weight**, directly from the provided raw text.
- **Shorthand & Jargon Recognition:** Utilizes NLP to interpret industry-specific abbreviations and handwritten-style text inputs, normalizing them into standardized trade terminology suitable for official records.
- **Automatic Data Cleaning & Formatting:** Transforms chaotic text descriptions into organized, structured fields (e.g., standardizing "Wt: 80kg" into a numerical weight value), ensuring the output is ready for direct database storage.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z.AI, a general language model, has been chosen for this system. Its strong core capability in multilingual comprehension, structured reasoning, and domain adaptation, which is critical for handling the complexity of global trade compliance. Z.AI's GLM demonstrates high proficiency in processing legal and regulatory text, enabling accurate summarization of dense government gazettes and policy updates into concise, actionable alerts. Its advanced contextual understanding allows it to interpret nuanced legal language across jurisdictions, making it well-suited for regulatory monitoring and law capture.

In addition, its capability in document analysis and cross-referencing supports intelligent document rectification and compliance validation workflows. It can efficiently compare structured and unstructured shipment data against dynamic regulatory requirements, detect inconsistencies, and identify missing documentation with high precision. The model's adaptability also enables

continuous learning from new regulations and trade patterns, ensuring the system remains up-to-date in a rapidly evolving global trade environment.

Furthermore, Z.AI's GLM offers scalable deployment and optimized performance for enterprise use, making it suitable for real-time processing and high-volume SME usage. Its balance of accuracy, efficiency, and multilingual support positions it as a reliable and future-ready choice for building an AI-powered compliance engine that minimizes risk and enhances cross-border trade operations.

4.3.2. **Prompting Strategy:** Our team has adopted a **Zero-Shot Structured Extraction** strategy. This approach involves providing the LLM with a clear set of instructions and a predefined output schema without providing specific labeled examples in the prompt.

We chose the **Zero-Shot** approach over Few-Shot or Multi-Step Agentic methods for the following reasons:

- **Handling High Input Variety:** Logistics text (emails, chat logs, shorthand notes) is extremely unpredictable. A Few-Shot approach (providing specific examples) might bias the model toward those specific formats. Zero-Shot leverages the LLM's broad pre-trained understanding to handle "any" messy text layout dynamically.
- **Latency & Efficiency:** As an SME tool, speed is critical. Multi-step agentic workflows (where the AI "thinks" through multiple steps or calls tools) introduce significant latency. Zero-Shot allows for **near-instant extraction**, ensuring a smooth user experience in the Streamlit UI.

  Structured Data for Database Integrity: By using a strict Schema-Constrained Prompt, we force the AI to return data in a consistent format (JSON). This is essential for our MariaDB backend, as it ensures the extracted Weight, HS Code, and Item Name can be saved directly into the 3NF tables without parsing errors.

4.3.3. **Context & Input Handling**: Our system is engineered to bridge the gap between volatile

human communication and rigid database structures. The handling logic ensures that even the messiest text is prepared correctly for the **FastAPI** backend and **MariaDB** storage.

- **Maximum Character Limit:** The system accepts up to **5,000 characters** per request.
- **Rationale:** This limit ensures the input stays well within the model's optimal context window, preventing "middle-of-the-document" forgetfulness and ensuring that crucial data like "Total Price" at the bottom of an invoice is never lost.

When the input text exceeds the 5,000-character threshold, the system follows a three-step protocol:

1. **Frontend Validation:** The **Streamlit** dashboard calculates the string length in real-time. If it exceeds the limit, the "Extract Information" button is disabled, and a warning prompts the user to "Divide the text into smaller sections."
2. **API Rejection:** The **FastAPI** endpoint includes a size-check middleware. Any payload exceeding the limit is rejected with a 413 Payload Too Large error, protecting the server from memory exhaustion.
3. **No Automatic Truncation:** To ensure **Data Integrity**, the system does *not* automatically cut off text. Truncating a logistics document might delete a vital HS Code or Weight value, leading to incorrect database entries. Instead, we require the user to provide complete, manageable segments.

For inputs that are within the size limit but are highly disorganized:

- **Preprocessing Layer:** The system uses a regex-based filter to strip out non-ASCII characters or excessive whitespace that doesn't contribute to data meaning, effectively "compressing" the input before it reaches the AI.

- **Contextual Anchoring:** The prompt instructs the AI to look for "Anchor Words" (e.g., *Invoice, To, Total, HS*) to orient itself within a messy layout, allowing it to ignore 90% of the conversational text while extracting 100% of the required data.

4.3.4. **Fallback & Failure Behavior:** To maintain the integrity of the logistics database and prevent the storage of inaccurate trade data, the system is designed with a "Safety-First" architecture. Rather than blindly accepting AI output, the system treats the model's response as a draft that must pass several validation layers.

The system addresses the risk of hallucination (AI making up data) through two primary mechanisms:

- **Structured Schema Validation:** The backend expects a specific JSON structure. If the model returns "off-topic" conversational text or a malformed JSON string, the system identifies this as a **Parsing Failure**. Instead of crashing, it catches the exception and prevents any data from being written to the database.
- **Confidence Thresholding & Null-Value Mapping:** The AI is instructed via the system prompt that if it cannot find a piece of information (e.g., a missing HS Code), it must return null or N/A rather than guessing. If the AI "hallucinates" a field that doesn't match the database data types (e.g., putting a name in a numeric weight field), the **SQLAlchemy** layer will reject the transaction.

If the first extraction attempt fails due to a timeout or a malformed response:

- **Automated Retry Loop:** The system is programmed to attempt a maximum of **two retries** with a slightly adjusted temperature setting. If the model still fails to produce a valid JSON output, the session status is updated to ERROR in the SessionState table (as seen in database.py).
- **Error Logging:** Every failure is captured in the error_log column of the database, providing developers with the raw trace of what went wrong for future model fine-tuning.

When the AI is unable to process a request, the system transitions into a **Graceful Error State** rather than leaving the user with a blank screen:

- **UI Alert System:** The Streamlit dashboard displays a localized error message (using the STATUS_MAPPING for EN/ZH/MS). For example, it might show *"Incomplete"* to signal that the AI couldn't find enough data to proceed.
- **The "Human-in-the-Loop" Path:** As demonstrated in the UI, all extraction results are shown in an editable preview format. If the AI provides an incorrect or "hallucinated" value, the **Human Escalation Path** allows the user to manually correct the field directly in the UI before clicking on a final "Confirm & Save" button. This ensures that a human always has the final word on what enters the permanent database.

5. User Stories & Use Cases

- **User Stories:** "As an SME business owner, I want to receive automated alerts about the latest customs law changes in simple language so that I can adjust my export strategy without hiring an expensive legal consultant."
- **Use Cases (Main Interactions):**

  Prompt to UI Flow: The system employs a **"Parse-Identify-Render"** pipeline to transform chaotic text into a structured 2D workspace.

Flow Stage	Action
Input	User pastes raw text (e.g., a messy invoice email). The system identifies the **Extraction Intent**, wrapping the input in a prompt that ignores "noise" and targets specific logistics entities.

Parsing	The AI extracts core trade components—**Item Name, HS Code, and Weight**—and structures them into a machine-readable JSON object via the FastAPI backend.
Rendering	The UI (Streamlit) receives the JSON and renders a **customizable 2D Canvas**. Each data point is mapped to a high-visibility, editable **Result Card** with a real-time status indicator (e.g., "Completed").
Finalization	The user reviews the rendered cards. Because the UI is interactive, users can manually correct or refine the AI's findings directly on the canvas before committing the final data to the **MariaDB** database.

Flow Generation: The system facilitates **Flow Generation** by transitioning validated data from a temporary extraction state into a permanent, structured record while maintaining semantic and visual consistency.

Stage	Action	Consistency Check
Trigger	User clicks "Confirm/Save".	Ensures human-in-the-loop validation.
Generation	System creates a new DB entry.	Maps extracted from JSON to SQL columns.
Maintenance	Tokens (Units/IDs) are preserved.	Prevents data corruption during transition.

6. Features Included (Scope Definition)

- **Text Analysis Engine:** Utilizes advanced Natural Language Processing (NLP) to perform structured data extraction from raw text inputs, allowing users to paste invoice details or shipping descriptions directly into the system for instant analysis.S
- **Structured Data Mapping:** Automatically identifies and categorizes key entities—such as HS Codes, item descriptions, and transaction values—from unformatted text strings provided by the user.
- **Multi-language Text Support:** Features built-in trilingual mapping for **English, Chinese, and Malay (EN/ZH/MS)** text, specifically optimized to interpret localized trade terminology and jargon used in the Malaysian market.
- **Input Validation & Cleaning:** Provides real-time feedback on the quality of the pasted text, flagging missing parameters or illogical data points before the compliance audit begins.

7. Features Not Included (Scope Control)

- Implementation of automated document filing and direct submission to government portals like Dagang Net
- Only for compliance auditing and diagnostic reporting, does not support physical logistics or warehouse management
- Multi-user collaborative document editing and real-time simultaneous data entry are not supported [Initial Versions]

8. Assumptions & Constraints

To ensure the sustainability and reliability of **Z-Clear**, we have defined specific boundaries regarding cost, technology, and performance. These constraints reflect the practical trade-offs required for an undergraduate-level AI project designed for the SME market.

- **LLM Cost Constraint:**
  - **Estimation:** Based on our current testing with the **ilmu-glm-5.1** model, we estimate a cost of approximately **0.02 to 0.05 USD per average user session** (assuming 1–2 extraction attempts and 1 auditing query).

- Cost-Saving Design Decision: We have implemented Schema-Strict Prompting to keep inference costs manageable. By forcing the model to output only raw JSON without conversational preambles or explanations, we significantly reduce the number of output tokens generated per request. Additionally, the system stores extracted data in MariaDB, so repeated views of the same shipment do not require re-running the LLM inference.

- **Technical Constraints:**
  - **Database Architecture:** The current system is optimized for **MariaDB (Vector)** and **SQLite**. While the backend is built on FastAPI, the integration with external ERP systems or older legacy customs software is not yet supported.
  - **File Format Limitations:** The current version is limited to processing **unstructured text input** via the UI. Direct uploads of PDF, JPG (OCR), or Excel files are currently out of scope; users must copy and paste text into the provided interface.
- **Performance Constraints:**
  - **Latency vs. Accuracy:** To ensure 100% data integrity for customs compliance, we prioritize "Model Thinking Time" over instant responses. Extraction typically takes **3–8 seconds**.
  - **Context Window Limit:** As noted in our input handling, there is a **5,000-character limit**. Processing larger datasets in a single prompt results in a significant drop in extraction accuracy and is therefore restricted to prevent data loss.
- **User Input:**
  - **Human-in-the-Loop Requirement:** The system is **not fully autonomous**. Because trade compliance carries legal weight, the model is unable to commit data to the permanent database without **explicit human approval**.
  - **Prompt Dependency:** The system cannot "guess" shipment details without a baseline text input from the user; it is a reactive middleware rather than a proactive document generator.

9. Risks & Questions Throughout Development

- **Extraction Accuracy:** To ensure the NLP does not misinterpret shorthand or informal layouts, we do not rely on simple keyword matching. The system utilizes **Contextual Anchoring**, where the model analyzes surrounding symbols (e.g., "$", "kg", "x") to determine the intent. Most importantly, the **Human-in-the-Loop** UI acts as the final fail-safe; every extraction is presented in an editable format for user verification before it is committed to the database.
- **Regulatory Latency:** Since we focus on checking against an existing database rather than live web-scraping, there is a risk of outdated rules. We mitigate this by including a **"Last Updated" timestamp** on all regulation-related outputs. If a retrieved rule is older than a specific threshold, the system flags the result with a warning, advising the user to verify against the latest government gazette.
- **Hallucination Risk:** To prevent the AI from generating non-existent HS Codes or tax rules, we use **Schema-Strict Constraint**. The model is explicitly instructed to return null if it cannot find a definitive match in the provided text. Additionally, the backend cross-references extracted codes against our **MariaDB Master Table**; if a code does not exist in the official registry, the system marks the entry as "Invalid" instead of blindly accepting the AI's output.
- **Data Security:** Protecting sensitive SME data is handled through **Environment Isolation** and **Transmission Minimization**. We utilize .env files for all API credentials and sensitive configurations. Furthermore, the system is designed to only send the necessary logistics text to the LLM API, keeping broader company identifiers (like bank details or private contact lists) stored locally within our encrypted database environment.

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